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Forecasting glucose values for patients with type 1 diabetes using heart rate data

Raffaele Giancotti [a], Pietro Bosoni [b], Patrizia Vizza [a,*], Giuseppe Tradigo [c], Agostino Gnasso [d], Pietro Hiram Guzzi [a], Riccardo Bellazzi [b], Concetta Irace [e], Pierangelo Veltri [f]

[a] Department of Surgical and Medical Sciences, Magna Graecia University of Catanzaro, Italy
[b] Department of Electrical, Computer and Biomedical Engineering, University of Pavia, Italy
[c] SMARTEST Lab., eCampus University, Novedrate, Como, Italy
[d] Department of Experimental and Clinical Medicine, Magna Graecia University of Catanzaro, Italy
[e] Department of Health Sciences, Magna Graecia University of Catanzaro, Italy
[f] Department of Computer Engineering, Modelling, Electronics and System, University of Calabria, Rende, Italy

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ABSTRACT

Background: Type 1 Diabetes Mellitus (T1DM) is a chronic metabolic disease affecting millions of people worldwide. T1DM requires patients to continuously monitor their blood glucose levels. Due to pancreatic dysfunctions, patients use insulin injections to correct glucose values by synthetic insulin. Continuous Glucose Monitoring (CGM) is a system which includes an algorithm allowing to measure (and in some cases to predict) glucose levels at a frequent sampling time. This enable implementing advanced devices, including automated insulin pump delivery. Nevertheless, CGM still presents some limitations, including (i) the delay (time lag) in detecting change in glucose levels compared to the traditional blood glucose measurement, and (ii) the lack of a sufficient and acceptable time to accurately predict glucose values.

Methods: We propose a framework based on a Gated Recurrent Unit (GRU) model to forecast both short- and long-term glucose values using heart rate (HR) and interstitial glucose (IG) values. The framework acquires HR and IG data and predicts glucose values with higher precision compared to state-of-the-art models. For training and testing the proposed framework, we used the OhioT1DM Dataset, which includes physiological data such as HR and IG values collected over an 8-week observation period. Additionally, we validated our framework using two other glucose datasets to ensure its generalizability across different HR and IG sampling frequencies. The proposed framework can be used to optimize the CGM system by incorporating patient HR measurements, thereby improving the prediction of short- and long-term glucose levels and reducing risks associated with conditions like hypoglycemia.

Results: Experimental tests were conducted using HR and IG data from the OhioT1DM Dataset, as well as from two additional T1DM patient datasets. We analyzed 6 patients from Ohio dataset while we validated the algorithm on 23 patients coming from two different university hospitals (6 from the University of Catanzaro medical hospital and 17 gathered from a validated study at IRCCS San Matteo Hospital in Pavia) for a total number of 29 patients. Our framework demonstrates an improvement in forecasting IG values in terms of RMSE and MAE for different choice of prediction horizons (PH). In the case of a PH of 5, 10, 20, 30, and 60 min, we reach an RMSE of 5.0, 9.38, 15.27, 20.48, and 34.16 respectively. The framework is freely available as an open-source, with an example dataset on a GitHub repository (see https://github.com/rafgia/attention_to_glycemia).

Conclusion: Our framework offers a promising solution for improving glucose level prediction and management in T1DM patients. By leveraging a GRU model and incorporating HR and IG values, we achieve more precise glucose level forecasting compared to state-of-the-art models. This approach not only enhances the accuracy of glucose predictions but also mitigates the risks associated with hypoglycemia.

* Corresponding author.
E-mail addresses: raffaele.giancotti@unicz.it (R. Giancotti), pietro.bosoni@unipv.it (P. Bosoni), vizzap@unicz.it (P. Vizza), giuseppe.tradigo@uniecampus.it (G. Tradigo), gnasso@unicz.it (A. Gnasso), hguzzi@unicz.it (P.H. Guzzi), riccardo.bellazzi@unipv.it (R. Bellazzi), irace@unicz.it (C. Irace), pierangelo.veltri@dimes.unical.it (P. Veltri).

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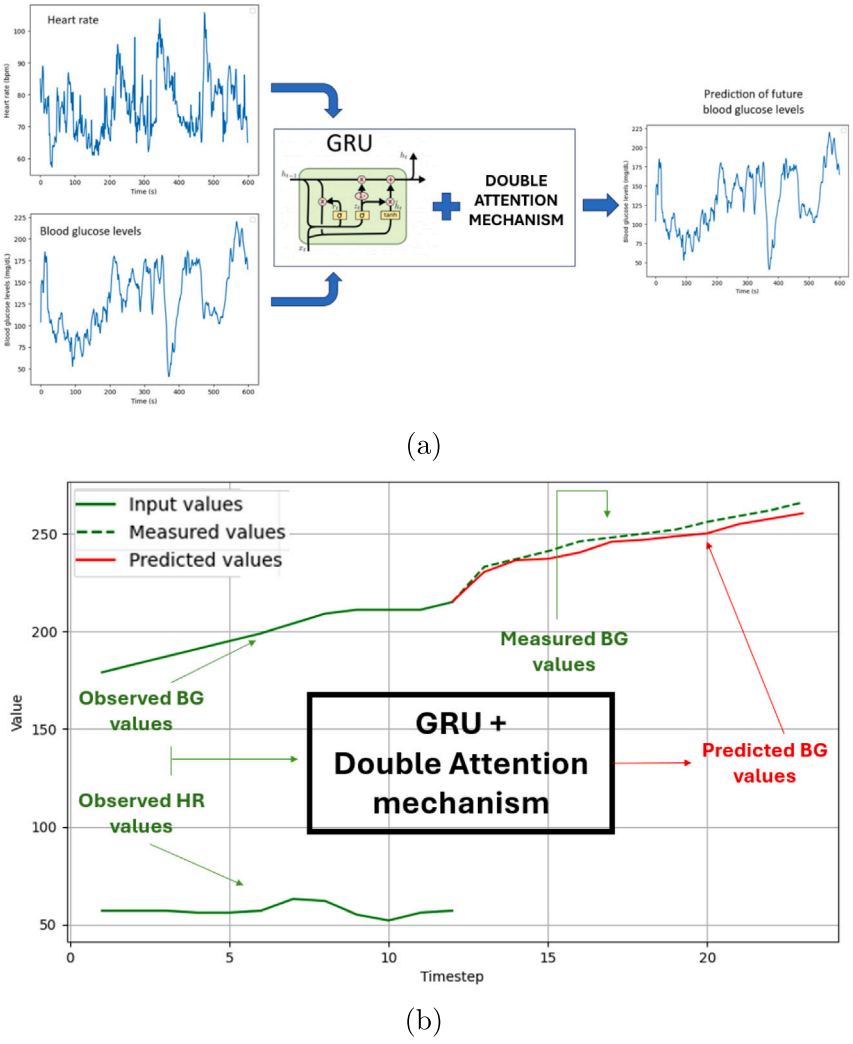


Fig. 1. Panel A illustrates the general architecture of the proposed framework, where interstitial glucose (IG) and heart rate (HR) values are used to train a Gated Recurrent Unit (GRU)-based model with an attention layer for glucose prediction. The framework takes as input both HR and IG signals, organized in a sequence of values collected over a specific time window (e.g., 12 values of HR and IG in a 1-hour time window). Panel B provides an example of the framework in action, showing IG and HR data in green and the predicted IG values in red.

1. Introduction

Type 1 Diabetes Mellitus (T1DM) is a chronic disease characterized by high levels of glucose in the blood due to the inability of the pancreas to produce insulin, the hormone that regulates interstitial glucose (IG) levels [1]. T1DM can lead to various complications for patients, including hyperglycemia, in which IG levels exceed 180 mg/dL, and hypoglycemia, in which IG levels can drop below 70 mg/dL. In severe cases, hypoglycemia can be life-threatening, emphasizing the need for precise glucose level control [2].

Continuous Glucose Monitoring (CGM) is a system that allows T1DM patients to monitor glucose and to self-manage insulin injections. It consists of a sensor that measures IG levels in the interstitial fluid rather than directly in the blood. The methodology is less invasive and more effective than the traditional blood glucose meter. Indeed, the sensor is applied to the patient's skin, continuously measuring IG levels. The CGM system can send alerts when these levels reach critical thresholds, and predict glucose levels over short time interval, thereby improving the quality of life for T1DM patients [3–6].

Despite its advantages, CGM still faces challenges, particularly in accurately measuring current IG levels (e.g., during the postprandial phase) and predicting future levels, which may be relevant for monitoring hypoglycemia events during sleep [7]. Accurate IG measurements are important to determine the correct insulin dosage. Integrating data from multiple physiological sensors, such as electrocardiograms and accelerometers, can enhance the reliability of the measurements obtained from a single glucose sensor [8]. In this paper, we provide a framework to improve the forecasting accuracy of CGM systems.

We focused on the relationship between heart rate (HR) and diabetes. In T1DM patients, hyperglycemia and hyperinsulinemia are associated with a high resting HR [9,10], thus that HR variability can be considered a useful trigger in T1DM patients. Elevated HR can be linked to higher glycated hemoglobin levels and increased insulin doses, while intensive management of diabetes is associated with a lower HR [11,12].

We propose a GRU-based framework, which offers several advantages such as better modeling of long-term dependencies and more efficient training in capturing significant features of the signals compared to conventional Recurrent Neural Network (RNN) variants. GRU-based models [13] have been successfully applied in various sequence modeling tasks, including natural language processing, time series forecasting, and biomedical data analysis. GRU models have also been used in different contexts such as finance [14] to predict financial time series [15–17] and in predicting climatic conditions. Thus, GRU can be considered a good candidate for accurate and efficient time series forecasting tasks [18]. We chose a GRU-based model because

preliminary tests showed that it met the expected values and results in clinical tests performed on a dataset from the University Magna Graecia Medical School. Our approach allows to capture and keep in memory temporal patterns even in previous time windows, using an attention mechanism that enhances model performance.

The proposed GRU-based model was trained and tested on a set of patient data extracted from [19] and also on the OhioT1DM Dataset[1] [20], demonstrating its ability to outperform the CGM system in a short-term interval. Overall, the proposed framework shows high accuracy in predicting IG levels, with clinically significant results that may enhance CGM systems' ability to predict hypoglycemic events in T1DM patients.

Fig. 1(a) illustrates the general architecture of this framework, where IG and HR values are used to train the GRU-based model with an attention layer for IG values forecasting. An example of how the proposed framework works is reported in Fig. 1(b) where data related to IG and HR levels are represented in green, while the IG predicted values in red.

The experimental results obtained using the proposed framework can be considered comparable to, and in some cases (e.g. very short time predictions) better than, those available in the literature. These findings suggest that in the glucose prediction challenge, there is still room for improvement in managing T1DM under various conditions.

1.1. Background

Machine learning (ML) models have been frequently used to predict IG values. In [21], ML algorithms and CGM data acquired in time-varying windows were used, combining inputs such as biosignals, carbohydrate intake, and insulin dosage. In another study [22], eXtreme Gradient Boosting (XGBoost) model was used to forecast IG levels with a 30-minute horizon, using various physiological signals recorded with wearable devices, including insulin infusion rates, HR, physical activity, galvanic skin response, and temperature. The XGBoost model achieved an RMSE of 23.48 in predicting future values. Additionally, in [23], the authors applied ML models to the Pima Indian Diabetes Dataset [24], focusing on feature engineering to improve glucose prediction. Although this study concerns type 2 diabetes patients, it is relevant due to the use of heterogeneous data sources to train the ML model. The Pima dataset was also employed in [25,26] where different yet interesting prediction models were proposed. In [27] the Seasonal Autoregressive Integrated Moving Average (SARIMA) was presented as a glucose prediction tool capable of forecasting glucose levels from limited data, performing comparably to more complex models.

Numerous studies have thus focused on Deep Learning (DL) models for predicting IG levels in T1DM patients, differing in both technical aspects and the input data used. In [28], a physiologically-informed Long Short-Term Memory (LSTM) network is used to forecast IG values based on carbohydrate intake and IG measurements. Similarly, in [19], the authors introduced a multi-patient and multivariate LSTM-based framework to develop a generalized IG prediction model, trained using data from a cohort of T1DM pediatric patients collected through the Freestyle Libre Flash Glucose Monitoring (FGM) system and Fitbit activity tracker. Another study [29] focused on developing a data-driven model using a Convolutional Recurrent Neural Network (CRNN) to predict IG levels in T1DM patient. In their experiments, as in ours, the OhioT1DM Dataset [20] was used, achieving glucose prediction over 30-minute and 60-minute horizons, with an Root Mean Squared Error (RMSE) of 17.45 mg/dL and 33.67 mg/dL, respectively.

In [30], authors use different machine learning models to predict the IG values with different PH (10, 20, 30, 40, 50, and 60 min). Results are then compared and results are reported by selecting the best performing one. We extracted the RMSE results on the different horizons. The results are comparable with those we report here. However, the paper treats a very large number of patients with respect to those used here on our GRU-based framework. Similarly, an important number of patient data has been used by a recurrent neural network model proposed in [31], where a cohort of more than 400 T1DM have been used with a large set of CGM data. The results are comparable in terms of RMSE metric, however, the used dataset is a very important and robust one with respect to the one we considered, used to validate the application of GRU-based enriched by attention method.

Our proposed framework is based on a Gated Recurrent Unit (GRU), which improves the prediction of future glucose values by capturing the temporal evolution of HR signals [32]. Our experimental results demonstrate that the GRU-based framework outperforms state-of-the-art methods, such as LSTM and CNN, in prediction tasks [33]. We also compared the glucose forecasting results with those from international glucose prediction challenges [34,35].

The integration of the Internet of Medical Things (IoMT) with ML and DL models can enhance continuous diabetes management by providing precise, personalized glucose level predictions. The methodology proposed here is novel in its use of a GRU model enriched with an attention mechanism, demonstrating that it is possible to improve glucose value forecasting performance. Finally, our framework aligns with the recommendations outlined in [36], which guide on using techniques, physiological data, and consensus-based methods to predict glucose values.

2. Methods

To predict BG values we designed a framework based on a GRU. We trained our architecture using a known and widely exploited dataset and we tested it with different sets of clinical data. This study adheres to the guidelines for Transparent Reporting of a Multivariable Prediction Model for Individual Prognosis or Diagnosis (TRIPOD) [37].

2.1. Functional architecture

The proposed GRU-based framework presents the general architecture reported in Fig. 1(a). The functional architecture presents a GRU model as the core of the proposed framework. Compared to traditional RNNs for time series prediction, GRUs perform well in capturing long-term dependencies within sequential data by selectively retaining crucial information from previous time steps. Furthermore, their reduced computational complexity and ease of implementation allowed their applications in many real-world scenarios consisting of large datasets and resource-constrained environments.

The GRU model used in the functional architecture is reported in Fig. 2. GRU presents two different gates: (i) the update gate, which regulates the integration of candidate activation vectors into new hidden states, and (ii) the reset gate, which determines the retention or omission of prior information. Comprising one or more layers, each GRU unit efficiently captures and retains relevant data from input sequences and network history. The hidden state of a GRU reflects accumulated information up to the present moment, continually updated with each time step. This improves the model by making it effective in time series prediction.

Using GRUs with IG and HR time series data, where current values are influenced by past ones, allows us to capture temporal dependencies, and model the dynamic nature of data. Moreover, the GRU can extract relevant features from both signals and integrate them to enhance prediction accuracy. This feature is particularly beneficial in our case, where the relationship between IG and HR levels can be complex and may vary over time. Furthermore, the flexibility and computational efficiency of GRU models compared to other types of RNNs make it a suitable choice for our task. Its ability to handle variable-length sequences efficiently ensures an effective modeling of the time

[1] The use of the OhioT1DM Dataset has been allowed to the DIMES, University of Calabria.

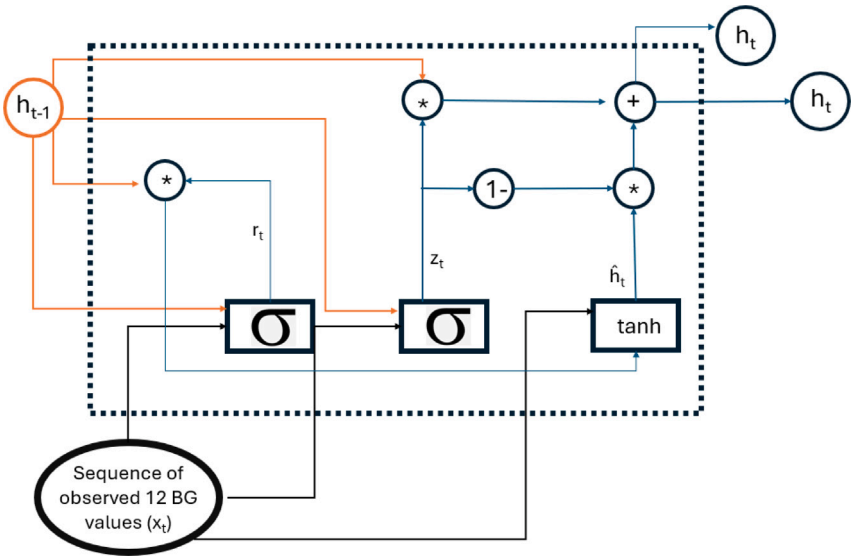


Fig. 2. Graphical representation of a GRU model [38]. At each timestep t, the GRU receives an input value x_t and the hidden state from the previous timestep h_{t-1}. Initially, at $t=0$, the previous hidden state is set to 0. The update gate $z(t)$ determines how much of the previous hidden state should be retained and how much of the new input should be considered as input to the sigmoid activation function (σ). The same activation function is used by the reset gate r_t to control how much of the previous hidden state should be combined with the new input. The new candidate memory content $\tilde{h}_t$ is calculated by using the hyperbolic tangent (*tanh*) activation function. The final hidden state is obtained by combining $\tilde{h}_t$ with the previous hidden state. The hidden state h_t is then used to predict the next value x_t. Lastly, this predicted value can be fed back into the model to perform further predictions.

series data. Additionally, the flexibility of the GRU architecture and its hyperparameters provides opportunities for optimization, leading to improved prediction accuracy.

Fig. 3 reports the functional architecture represented as a workflow diagram. The Figure reports processes involving signals used to extract the most relevant features. Two GRU layers are used, one for HR data and one for IG data (both x_t). Reading from the top part to the bottom one of Fig. 3 the two input data sources flow through the workflow in the two paths. The update gate (z_t) determines how much of the previous hidden state (h_{t-1}) should be retained and how much of the input information should be included in the current hidden state. Similarly, the reset gate (r_t) controls how much of the previous hidden state should be ignored when computing the current hidden state. Both gates are computed through a sigmoid activation function (e.g., $\sigma(x) = \frac{1}{1+e^{-x}}$). A candidate hidden state (h'_t) is then evaluated (see (iii) in Figure) using the input features and the Reset Gate. The Candidate Hidden state represents the new information that can be included in the Current Hidden state. The Current Hidden state (h_t) is, then updated, based on the update gate, incorporating information from the previous hidden state and the Candidate Hidden state.

To improve the time series predictions, we enrich the GRU framework by integrating an *attention mechanism* [39–41], which dynamically prioritizes specific segments of the input sequence by assigning greater significance to certain time events during the process. The attention mechanism assigns weights to a single element within the input sequence, enabling the model to highlight relevant information. A context vector c_t based on the current hidden state h_t as well as the sequence of previously hidden state values are estimated using the attention mechanism. The context vector is assessed through the allocation of specific weights to each previous hidden state h_i w.r.t. its relevance to the current hidden state h_i. A scoring function is used to calculate these weights by measuring the similarity or compatibility between h_t and h_i. By combining the context vector c_t with the current hidden state h_t the functional architecture generates an adjusted (value) for the hidden state variable.

Thus, it is possible to obtain prediction values that are more reliable considering the relevant parts of the input sequence. A temporal attention mechanism is applied to the sequence of predicted values (output). It allows to tune the importance of each time step dynamically to improve performances.

The output of each step is used as input to a hyperbolic tangent activation function *tanh*. The activation module is performed by means of a softmax activation function. This phase allows to use hidden status of the GRU to improve prediction performances. The attention-weighted sum of the GRU outputs for IG and HR data is used as input for an additional layer (i.e., Lambda layer [42], see the bottom part of the functional architecture figure). Finally, the attention-weighted outputs are concatenated to obtain the final concatenated attention-weighted outputs, then used by a final step of the dense layer to produce the IG prediction values. Additional information can be found at the GitHub repository hosting code and data samples (https://github.com/rafgia/attention-to-glycemia).

The attention layer consists of three matrices: a query matrix, a key matrix, and a value matrix. The query matrix represents the current hidden state of the GRU and serves as the reference point for attention computation. The key matrix captures information from each time step of the input sequence, while the value matrix holds the corresponding values. Through a set of trainable parameters, the attention mechanism calculates attention scores, indicating the importance of each time step to the prediction task at hand. These scores are then used to obtain a weighted result, generating a context vector that encapsulates relevant information for the prediction.

Fig. 3 shows the model and how it temporally relates to HR and IG levels obtaining the prediction horizon (PH) capabilities.

Finally, the HR variation in T1DM patients is used as a starting point as well in other studies such as in [43]. One of the aims of our contribution regards the possibility of predicting the glucose level variation and relating it to the insulin administration process.

2.2. Data

For the training and testing phases of our GRU-based model, we used the OhioT1DM Dataset. For each patient, we considered BG and HR signals during a large time window (one month on average). The OhioT1DM Dataset encompasses eight weeks of data for 12 individuals (5 female and 7 male, average age 50 ±10 years) affected by T1DM. The

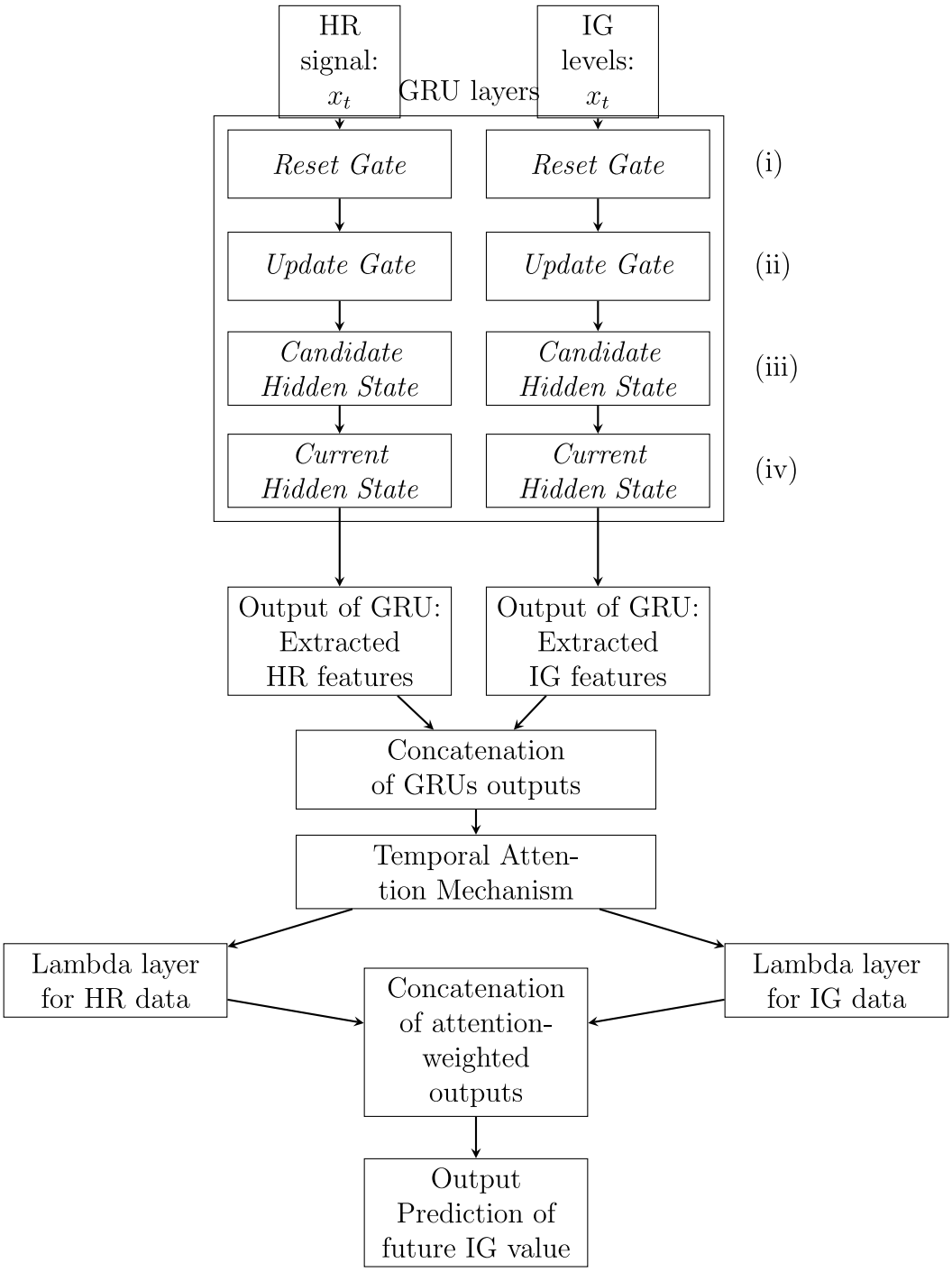


Fig. 3. Workflow used by the GRU-based framework for predicting IG values. Input signals pass through two parallel GRU layers, which extract features from both time series (steps (i), (ii), (iii), and (iv)). These features are concatenated, and a temporal attention mechanism is applied to dynamically weigh each time step. Finally, Lambda layers compute the attention-weighted sum for IG and HR data separately, which are then concatenated to generate the final IG value prediction.

dataset is anonymized. Throughout the data collection period, all participants used insulin pump therapy combined with CGM. Specifically, they wore either Medtronic 530G or 630G insulin pumps and utilized Medtronic Enlite CGM sensors. Data contributors recorded life-event information via a customized smartphone application and acquired physiological data from a fitness band. The OhioT1DM Dataset includes different types of signals, such as (i) CGM interstitial glucose values acquired every 5 min; (ii) blood glucose values acquired through self-monitoring (finger sticks); (iii) insulin doses, both bolus and basal; (iv) self-reported meal times with carbohydrate estimates; (v) self-reported times of exercise, sleep, work, stress, and illness; (vi) data from the Basis Peak or Empatica Embrace band. The *Basis heart rate* data is only available for people who wore the Basis Peak sensor band. Thus the processing phase involved 6 out of 12 OhioT1DM Dataset patients. HR data is recorded every minute and then averaged over 5 values to have a time correspondence with each IG value, which is measured every 5 min. The dataset is organized in two folders, one containing training data and the other test data. For the task, we chose to use only the IG levels and HR values while maintaining the division between training and testing already provided by the OhioT1DM Dataset.

We then tested our framework on data coming from 29 patients. 6 patients have been collected along one year of testing from the

Table 1
Comparison of IG value forecasting using different models on the OhioT1DM dataset. All models considered take past IG and heart rate (HR) values as input. The lowest RMSE and MAE, both measured in mg/dL, for each prediction time window are highlighted. The proposed GRU+DA model performs well for predictions 5, 15, and 20 min ahead. For predictions 10 min ahead, the GRU+A model outperforms the others.

	5 min ahead			10 min ahead			15 min ahead			20 min ahead		
	Loss	RMSE	MAE	Loss	RMSE	MAE	Loss	RMSE	MAE	Loss	RMSE	MAE
LSTM	40.48	6.14	3.32	110.19	9.35	5.88	206.47	12.54	8.28	315.12	16.56	11.13
GRU	45.23	6.04	3.57	116.46	9.68	6.08	212.32	13.40	8.58	326.47	15.88	10.69
Gru+A	42.68	5.27	3.01	113.66	**9.16**	**5.55**	213.31	12.86	8.08	321.49	15.56	10.45
GRU+DA	40.39	**5.00**	**2.73**	111.63	9.38	6.10	209.10	**12.39**	**7.96**	312.24	**15.27**	**10.07**

Table 2
Results of forecasting task of BG values with different PH (5, 10, 15, 20, 30 and 60 min) in terms of RMSE, MAE, R-squared and MAPE metrics.

	GRU+DA			
	RMSE	MAE	R-squared	MAPE
5 min ahead	5.00	2.73	0.9924	2.31%
10 min ahead	9.38	6.10	0.9776	4.01%
15 min ahead	12.39	7.96	0.9571	5.71%
20 min ahead	15.27	10.07	0.9328	7.30%
30 min ahead	20.48	14.59	0.8865	9.57%
60 min ahead	34.16	25.54	0.6992	16.91%

University of Catanzaro Medical Hospital, while 17 have been gathered from a validated study at IRCCS San Matteo Hospital in Pavia.

2.3. Model training and metrics

The proposed framework acquires data by using a time window of varying dimensions (e.g., 1 h or 2 h). For example, while using the OhioT1DM Dataset, the time window is equal to one hour, sampling 12 data every 5 min. For instance, let t be the acquisition time (i.e. 1 h for the OhioT1DM Dataset), the PH, future value, is obtained by using the values acquired in t. The t+PH value is predicted by guessing the next upcoming 5-minute IG value. The model moves forward the time window acquiring the measured IG value and HR value, still maintaining a fixed time window (e.g., 12 values for the OhioT1DM Dataset). Obviously while acquiring, the latest time values are discharged using a moving window. The model is made up of 64 units, and 20 epochs with a batch size of 32. To train the network, we used the *Adam* optimizer [44] for its adaptive learning rate and its capability to accelerate the convergence during training when long time series are used. The model is geared to the Mean Squared Error (MSE) loss function, which is suitable for regression tasks.

To facilitate comparison and assess the effectiveness of our model against others, we selected several architectures: *(i)* a simple LSTM model, *(ii)* a simple GRU model, and *(iii)* a GRU model followed by a single layer of attention (GRU+A), which captures information from both IG and HR signals. Finally, the GRU-based framework (GRU+Double Attention) has been tested with different time windows of 5, 10, 15, and 20 min, respectively.

RMSE and MAE have been used as performance metrics. We also computed: (i) R-squared index, which determines the proportion of variance in a dependent variable that can be predicted or explained by an independent variable, and (ii) Mean Absolute Percentage Error (MAPE) which expresses the error as a percentage of the actual values. Both indexes are defined as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{1}$$

$$\text{MAPE} = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \times 100 \tag{2}$$

where:

- y_i is the actual value at time i,
- $\hat{y}_i$ is the predicted value at time i,
- $\bar{y}$ is the mean of the actual values,
- n is the total number of observations.

Such indexes have been used to evaluate the prediction performances to measure the PH for time-window of prediction, respectively 30 and 60 min ahead.

3. Results and discussion

To prove the relevance of HR values combined with IG ones, we tested the framework with a simple GRU using only IG values. We also performed tests considering the GRU plus attention mechanism using only IG values. Experimental tests were carried on using a PC equipped with an Intel(R) Core(TM) i7-10750H CPU model with a base frequency of 2.60 GHz (up to 5.00 GHz, 6 cores, 12 threads), RAM with a capacity of 16 GB, type DDR4 with a speed of 2933 MHz. The laptop is also equipped with an NVIDIA GeForce GTX 1650 Ti GPU with 4 GB of dedicated GDDR6 memory.

The results of forecasting future IG values with the GRU-based framework+Double Attention (GRU+DA) are reported in Table 1, displaying RMSE and Mean Absolute Error (MAE). Additionally, the training loss is reported. The Table reports forecasting results obtained using different models on the OhioT1DM Dataset. Note that the proposed framework outperforms the other approaches in terms of the lowest RMSE and MAE values for the prediction of upcoming IG values in 5, 15, and 20 min ahead. In case of 10 min ahead, the proposed GRU+A performs better than the framework using double attention. Please note that the double attention applied a double attention layer in the latter part of the functional architecture (see Fig. 3).

In the Table 2 we report the test conducted on a larger time-window of prediction, respectively 30 and 60 min ahead. For all the considered PH, we report all the metrics we considered, i.e. RMSE, MAE, R-squared, and MAPE (see Section 2).

Table 3 shows how using a GRU with only past IG values as input, the RMSE and MAE estimated are worse than those obtained using a combination of HR and IG values. Moreover using the GRU enriched with the attention mechanism both RMSE and MAE values are higher than those obtained by the other strategies. This confirms the benefits of using our model and the effectiveness of incorporating past HR values together with IG values.

To validate the clinical significance of the results, we used the Clarke's function [45]. The Clarke Error Grid Analysis (CEGA) allows to evaluate the results obtained from the prediction of IG levels. CEGA, shown in Fig. 4, is a pivotal tool for evaluating the accuracy of CGM systems and predictive models. Widely used in diabetes management, CEGA provides valuable insights into the clinical accuracy of glucose prediction models. CEGA partitions the glucose prediction space into five distinct zones: (i) *zone A* where the predictions are considered clinically accurate; (ii) *zone B* represents predictions that, while clinically acceptable, may not prompt the most appropriate clinical action; (iii) *zone C* describes predictions leading to benign clinical errors; (iv) *zone D* indicates predictions resulting in moderate to severe clinical errors and (v) *zone E* indicates predicted values which are not clinically reliable (e.g. false negatives). By integrating the standard metrics (MAE and RMSE) alongside the Clarke Error Grid (CEG), we aim to provide a more robust assessment of the predictive performances, ensuring

Table 3
Results obtained by using the OhioT1DM Dataset as input for the proposed GRU-based framework. Tests have been performed to forecast future IG values respectively 5, 10, 15, and 20 min using a simple GRU and a GRU with an attention mechanism. The Table compares the performance of the proposed GRU-based framework when using a time window with only previous IG values and when using IG values enriched with HR ones.

Exp. Setup	without HR			with HR		
	Loss	RMSE	MAE	Loss	RMSE	MAE
GRU						
5 min ahead	42.83	5.44 mg/dL	2.91 mg/dL	45.23	6.04 mg/dL	3.57 mg/dL
10 min ahead	123.04	9.96 mg/dL	5.83 mg/dL	116.46	9.68 mg/dL	6.08 mg/dL
15 min ahead	210.83	12.99 mg/dL	8.40 mg/dL	212.32	13.40 mg/dL	8.58 mg/dL
20 min ahead	432.97	18.31 mg/dL	12.53 mg/dL	326.47	15.88 mg/dL	10.69 mg/dL
GRU+DA						
5 min ahead	59.68	7.02 mg/dL	5.33 mg/dL	42.68	5.27 mg/dL	3.01 mg/dL
10 min ahead	147.35	9.98 mg/dL	6.30 mg/dL	113.66	9.16 mg/dL	5.55 mg/dL
15 minutes	239.12	14.76 mg/dL	10.08 mg/dL	213.31	12.86 mg/dL	8.08 mg/dL
20 min	372.27	17.69 mg/dL	12.40 mg/dL	321.49	15.56 mg/dL	10.45 mg/dL

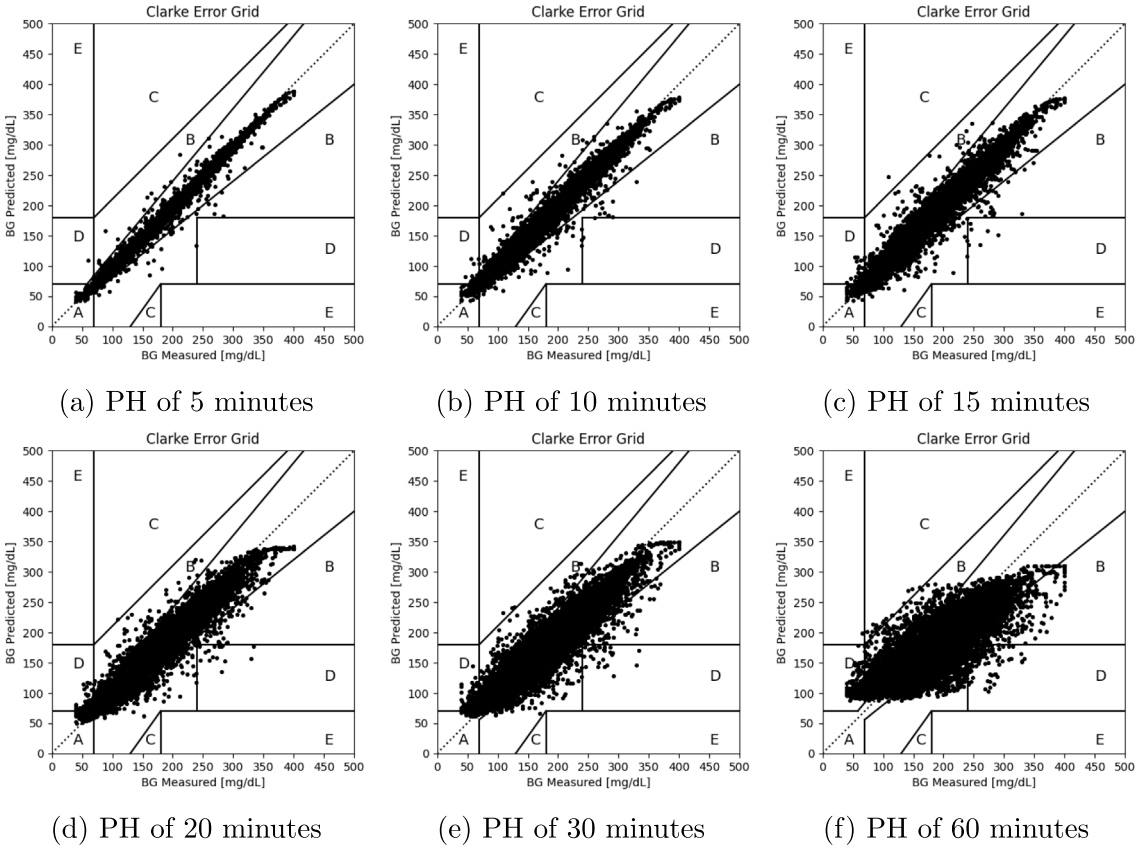


Fig. 4. The Clarke Error Grid Analysis (CEGA) is reported for all of PH considered in the forecasting task of IG values, where each prediction is set against the reference value on a scatterplot. This graphical representation delineates the zones, enabling clinicians to quickly assess the accuracy and safety of their predictions. The grids show how the GRU-based framework is used to predict IG values is precise for 5 min in advance, and at the same time present acceptable results using longer PH.

that our results are both clinically meaningful and consistent with established practices in glucose prediction research.

To confirm the positive results obtained in the above described experiment (shown in Table 1), we applied our tool to the dataset used in [19] (denoted as PV-dataset). The PV-dataset consists of clinical data gathered from a cohort of patients diagnosed with T1DM. Data consist of 17 adolescents (9 female and 8 male, average age 12.30 ±4.33 years) recruited from the Pediatric Diabetology Unit at Fondazione IRCCS Policlinico San Matteo Hospital in Pavia, Italy. Data contain IG levels recorded every 15 min with a Flash Glucose Monitoring (FGM) system (Abbott FreeStyle Libre, Alameda, CA, USA), resulting in 96 automatic measurements stored daily. HR measures with a sampling frequency of one minute are also provided by a Personal Fitness Tracker (PFT) (FitBit Charge HR, San Francisco, CA, USA). The dataset contains a monitoring time window of 200 days, gathering simultaneous glucose and HR monitoring. 80% of the data has been used for training and 20% for testing. In the latter case, since IG and HR values are recorded every 15 min, for the training phase we take sequences of 8 values, corresponding to 2 h of measurements, and predict IG values, respectively, 15, 30, 45, and 60 min in advance. Differently from the OhioT1DM Dataset where data have an acquisition rate of 5 min, the PV-dataset presents a data acquisition rate of 15 min. Consequently, the proposed model's predicted PH should be tuned according to the data acquisition rates. Table 4 reports the results of applying our GRU-based framework on the PV-dataset. The Table shows RMSE and MAE results of predicting IG values in different time windows.

Table 4
Results measured in terms of RMSE and MAE describing the reliability of predicting IG values obtained using the proposed framework on the PV-dataset. Each line indicates different time windows. IG and HR values have been acquired every 15 min. The training phase has been performed using a sequence of 8 values, corresponding to 2 h of measurements. Forecasting future IG values at 15, 30, 45, and 60 min in advance, respectively are also reported.

Validation of framework with PV-dataset			
Exp. setup	Loss	RMSE	MAE
15 min ahead	79.92	5.57 mg/dL	3.41 mg/dL
30 min ahead	342.29	9.30 mg/dL	5.83 mg/dL
45 min ahead	729.34	13.28 mg/dL	8.31 mg/dL
60 min ahead	1156.06	16.15 mg/dL	11.16 mg/dL

Table 5
RMSE measured by using GRU-based framework as compared with the performance of the LSTM model. Both methods predict the IG value with a time horizon of 15 min in advance, the first based on the previous 45 min acquired values for IG and HR, and the second using sequences of values of the previous 120 min. The RMSE value obtained by using LSTM is higher with respect to the one obtained by using the here proposed model.

Sequence used in input	Model	RMSE
IG and HR previous values	LSTM	10.92 mg/dL
IG and HR previous values	GRU+Double attention	5.57 mg/dL

Table 6
Table reports the performance of the proposed GRU-based model measured on the CZ-dataset in terms of RMSE and MAE of predicting future IG values 5, 10, 15, and 20 min. IG and HR values have been acquired with a sampling frequency of 5 min. We take sequences of 12 measured IG and HR values, corresponding to 1 h of measurements, and predict future IG values.

Validation of framework with CZ-dataset		
Exp. setup	RMSE	MAE
5 min ahead	6.08 mg/dL	3.79 mg/dL
10 min ahead	11.07 mg/dL	6.46 mg/dL
15 min ahead	14.07 mg/dL	9.17 mg/dL
20 min ahead	17.58 mg/dL	12.26 mg/dL

Moreover, we compared the results reported in Table 4 with those obtained by using LSTM on the same dataset [19]. Table 5 reports the comparison between our results with those obtained, in terms of RMSE of forecasting IG values 15 min ahead.

The GRU-based framework has been validated also using a dataset from the University Hospital of Magna Graecia University, which we indicate as CZ-dataset. This dataset contains data coming from 6 patients (3 females and 3 males, average age 45.16 ±10.24 years) acquired by using wearable devices in approximately 7 days. Each acquisition has been performed using a Dexcom G6 glucose meter with a frequency of 5 min, while heart rate has been recorded via a Polar Vantage M fitness tracker, with a frequency of 1 min. Data have been pre-processed to uniform sampling frequency among acquisitions. The average heart rate values for every 5 data points have been evaluated. As in the general purpose case, we train the GRU model using the OhioT1DM Dataset. Prediction results for IG values are reported in Table 6, both in terms of RMSE and MAE.

The GRU-based framework considers one hour of data to forecast IG levels 5, 10, 15, and 20 min ahead. Our results demonstrate a significant enhancement in prediction accuracy, as shown by the improved RMSE and MAE metrics with respect to existing methodologies. Our approach also confirmed the benefit of using the HR values as an additional input to the prediction model, could accurately predict hypoglycemia cases, particularly during sleep night time. This can be considered a substantial step towards more effective diabetes management strategies.

Finally,we compared the results we obtained while using the OhioT1DM Dataset (both 2018 and the updated one acquired in 2020), with results of the glucose prediction challenge 2020. Artificial neural network algorithms were developed using the OhioT1DM Dataset, including LSTM, Bidirectional LSTM (BiLSTM), Convolutional LSTMs, and Temporal Convolutional Network (TCN). Sequence-to-sequence models were created and evaluated on a test dataset with 6 unique subjects. The sequence-to-sequence BiLSTM model achieved the lowest RMSE for 30-minute predictions (RMSE mean of 27.4), highlighting the potential of neural networks in managing Type 1 diabetes. Our models reach an RMSE mean of 21.21. Thus, by comparing the reported results, we claim that our method still has room for improvements with respect to the available ones, as also reported on the application on available real datasets.

Of course, thee proposed GRU-based framework has some limitations, which could be addressed in future research. A first area of improvement regards the management of missing values. In future investigations we plan to estimate missing values by resorting to model-based strategy made possible by the integration of different data sources, i.e. coupling IG, HR, and wearable signals, say accelerometers. Moreover, future developments may concern the customization of the proposed framework, with the design of a wearable prototype for the acquisition and processing of the signals and an on-field assessment for a user-centric solution.

4. Conclusion

A GRU-based framework to forecast IG values by using physiological signals has been presented. By incorporating HR data alongside IG measurements, the framework has shown to state-of-art prediction accuracy. In particular, the results were compared with other available models and with the outcomes of international challenges performed on the OhioT1DM Dataset, demonstrating competitive performance in terms of RMSE. Future directions include enhancing the architecture by enriching the data to improve accuracy, particularly in cases of missing data. Moreover, the aim is to integrate the framework in a general-purpose architecture that includes an alert mechanism for nighttime glucose monitoring.

CRediT authorship contribution statement

Raffaele Giancotti: Writing – original draft, Software, Resources, Methodology, Funding acquisition, Data curation. **Pietro Bosoni:** Resources. **Patrizia Vizza:** Visualization, Supervision. **Giuseppe Tradigo:** Visualization, Supervision. **Agostino Gnasso:** Investigation, Formal analysis, Conceptualization. **Pietro Hiram Guzzi:** Writing – original draft, Supervision, Software, Methodology. **Riccardo Bellazzi:** Supervision. **Concetta Irace:** Supervision, Investigation, Conceptualization. **Pierangelo Veltri:** Writing – original draft, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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